

JEE MAIN REVISION MASTER: POLYMERS OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 79 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Polymers — Standard Calculations & Problem Archetypes	35 PYQs	Quantitative Analysis
Very High	Polymers — Applied Cases, Reactions & Traps	24 PYQs	Applied Topics
Very High	Polymers — Core Principles & Governing Laws	23 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Polymers — Standard Calculations & Problem Archetypes	$AlEt_3 + TiCl_4 \rightleftharpoons Al(Et)_3 \rightleftharpoons n = \frac{963}{107} = 9$
Polymers — Applied Cases, Reactions & Traps	$TiCl_4, SnCl_4, AlCl_3, BF_3$
Polymers — Core Principles & Governing Laws	$CF_2 = CF_2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Polymers — Standard Calculations & Problem Archetypes (Very High Yield — 35 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Polymers.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "Match List I with List II : List I - (Monomer) List II - (Polymer) (A) Tetrafluoroethene (I) Orlon (B) Acrylonitrile (II) Natural rubber (C)..." [Ans: A]
- [2023 Morning Shift] Q#1 (MCQ): "Polymer used in orlon is :..." [Ans: N/A]
- [2023 Evening Shift] Q#2 (MCQ): "Match List I with List II LIST I LIST II A. Weak intermolecular forces of attraction I. Hexamethylenediamine + adipic acid B. Hydrogen bondi..." [Ans: N/A]

3.2 Polymers — Applied Cases, Reactions & Traps (Very High Yield — 24 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Polymers.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2023 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I : Ethene at 333 to 343K and 6 - 7 atm pressure in the presence of $AlEt_3$ and $TiCl_4$..." [Ans: N/A]
- [2023 Morning Shift] Q#2 (MCQ): "The polymer X - consists of linear molecules and is closely packed. It is prepared in the presence of triethylaluminium and titanium tetra..." [Ans: N/A]
- [2023 Morning Shift] Q#2 (MCQ): "Caprolactam when heated at high temperature in presence of water, gives..." [Ans: N/A]

3.3 Polymers — Core Principles & Governing Laws (Very High Yield — 23 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Polymers.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Polymers.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Polymers PYQs.

Representative Source PYQs:

- [2023 Morning Shift] Q#3 (MCQ): "Match the following Column A Column B (a) Nylon 6 (I) Natural Rubber (b) Vulcanized Rubber (II) Cross Linked (c) cis-1, 4-polyisoprene (III)..." [Ans: A]
- [2023 Evening Shift] Q#3 (MCQ): "Buna-S can be represented as:..." [Ans: N/A]
- [2023 Evening Shift] Q#3 (MCQ): "Match List I with List II List I List II A. Elastomeric polymer I. Urea formaldehyde resin B. Fibre Polymer II. Polystyrene C. Thermosetting..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 15th April Morning Shift (Q#1)

Match List I with List II : List I - (Monomer) List II - (Polymer) (A) Tetrafluoroethene (I) Orlon (B) Acrylonitrile (II) Natural rubber (C) Caprolactam (III) Teflon (D) Isoprene (IV) Nylon-6 Choose the correct answer from the options given below :

- (A) (A)-(III), (B)-(I), (C)-(IV), (D)-(II) **[Correct]**
- (B) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
- (C) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)
- (D) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Correct Answer: A

Solution Analysis: The correct match between List I (Monomer) and List II (Polymer) is as follows: (A) Tetrafluoroethene → (III) Teflon (B) Acrylonitrile → (I) Orlon (C) Caprolactam → (IV) Nylon-6 (D) Isoprene → (II) Natural rubber So, the correct answer is: A-III, B-I, C-IV, D-II

Benchmark Problem 2: JEE Main 2023 (Online) 11th April Evening Shift (Q#1)

Given below are two statements : Statement I : Ethene at 333 to 343K and 6 - 7 atm pressure in the presence of AlEt_3 (3) and TiCl_4 (4) undergoes addition polymerization to give LDP. Statement II : Caprolactam at 533 - 543K in H_2O through step growth polymerizes to give Nylon 6. In the light of the above statements, choose the correct answer from the options given below:

- (A) Both Statement I and Statement II are true
- (B) Statement I is true but Statement II is false
- (C) Statement I is false but Statement II is true
- (D) Both Statement I and Statement II are false

Correct Answer: None

Solution Analysis: Statement I is false. Ethene (ethylene) under certain conditions (high pressure, high temperature) in the presence of catalysts does undergo addition polymerization, but it forms polyethylene, not LDP (Low-Density Polyethylene) specifically. The formation of low-density polyethylene (LDPE) vs high-density polyethylene (HDPE) depends on the specific conditions and catalysts used. The catalysts mentioned (AlEt_3 and TiCl_4) are associated with the production of HDPE through the Ziegler-Natta process, not LDPE. Statement II is true. Caprolactam polymerizes to form Nylon 6 via a ring-opening polymerization.

Benchmark Problem 3: JEE Main 2023 (Online) 13th April Morning Shift (Q#3)

Match the following Column A Column B (a) Nylon 6 (I) Natural Rubber (b) Vulcanized Rubber (II) Cross Linked (c) cis-1,4-polyisoprene (III) Caprolactam (d) Polychloroprene (IV) Neoprene Choose the correct answer from options given below:

- (A) a → III, b → II, c → I, d → IV **[Correct]**
- (B) a → II, b → III, c → IV, d → I
- (C) a → IV, b → III, c → I, d → II
- (D) a → III, b → IV, c → I, d → II

Correct Answer: A

Solution Analysis: Nylon 6 is a synthetic polymer made from caprolactam, so it corresponds to Column A, option (c), which matches with Column B, option (I). Vulcanized rubber is rubber that has been treated with sulfur or other additives to make it more durable, so it corresponds to Column A, option (b), which matches with Column B, option (II). Cis-1,4-polyisoprene is a naturally occurring polymer found in natural rubber, so it corresponds to Column A, option (a), which matches with Column B, option (III). Polychloroprene is a synthetic polymer used to make neoprene, a type of synthetic rubber, so it corresponds to Column A, option (d), which matches with Column B, option (IV).